Philosophy of Space and Time

Lecture 11: *Special Relativity II*

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Today's Plan

1. Recap
2. Light Cones and Causal Order
3. Interval
4. The Speed of Light?
5. What Clocks Measure
6. The Twins
7. Closing

Where We Left Off

Before the exam we introduced **Minkowski spacetime** as the next step in understanding space and time:

- Galilean spacetime still contains *absolute simultaneity*
- Einstein's two postulates (relativity + constancy of c) force its abandonment
- Result: a new geometry of **spacetime** with light cones replacing simultaneity slices

Today: further consequences of this geometry, understanding the spacetime interval.

The Two Postulates

Principle of Relativity

The laws of physics are the same in every inertial frame.

Constancy of the Speed of Light

In every inertial frame, light propagates in vacuum at the same speed c, independent of the motion of source or observer.

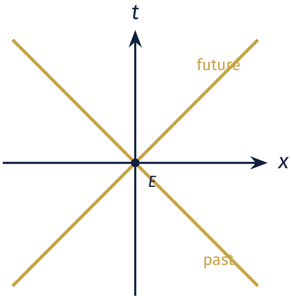
These two principles are jointly inconsistent with the Galilean addition of velocities. Something has to give—and Einstein argues that what gives is **absolute simultaneity**.

Light Cones
and Causal Order

The Light Cone

At each event E, light propagates outward in all directions at speed c. In a spacetime diagram, this traces out a cone.

- **Future light cone**: events reachable from E by a light signal
- **Past light cone**: events from which a light signal could have reached E
- The light cone is the same for all observers



Three Kinds of Separation

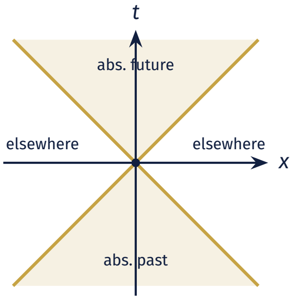
Two events E_1 and E_2 can be related in one of three ways, depending on whether E_2 falls inside, on, or outside the light cone of E_1.

- **Timelike**: E_2 inside E_1's light cone. E_1 can causally influence E_2 (or vice versa).
- **Lightlike (null)**: E_2 *on* E_1's light cone. Connectable by a light signal.
- **Spacelike**: E_2 outside E_1's light cone. Neither can influence the other.

These distinctions are *absolute*—they hold for all observers. Unlike simultaneity, they are part of the geometric structure of Minkowski spacetime itself.

The Causal Structure

- Events in the **absolute future** of *E*: causally accessible *from E*
- Events in the **absolute past** of *E*: could causally influence *E*
- Events in the **elsewhere**: neither in *E*'s past nor its future
- For spacelike-separated events, even the time-order is frame-dependent



The Invariant Interval

Although time and distance separately are frame-dependent, the combination is not:

$$\Delta s^2 \;=\; (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - c^2(\Delta t)^2$$

- $\Delta s^2 < 0$: timelike (events on a possible worldline)
- $\Delta s^2 = 0$: lightlike
- $\Delta s^2 > 0$: spacelike

?? *The interval is the genuinely geometric quantity in Minkowski spacetime—the analog of distance in Euclidean space, but with a minus sign for the time term.*

The Spacetime Interval

Geroch's Construction of the Interval

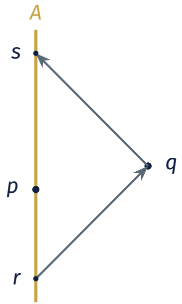
Geroch defines the interval not by writing down coordinates, but from the operational behavior of clocks and light signals.

For events p and q, take any inertial observer A through p, with A's clock set to zero at p. Two further events r, s, and times for A defined as follows:

- t_1: $t(s) - t(p)$ (positive for this case)
- t_2: $t(p) - t(r)$ (positive for this case)

Geroch's **interval**:

$$I(p, q) = t_1 \cdot t_2 .$$



What All Observers Agree On

In coordinates relative to A, with q at $(\Delta t, \Delta x)$:

$$t_1 = \Delta t + \frac{\Delta x}{c}, \qquad t_2 = \frac{\Delta x}{c} - \Delta t, \qquad I(p,q) = t_1 t_2 = \frac{(\Delta x)^2}{c^2} - (\Delta t)^2.$$

(Note: I have corrected a mistake on the early slides, which had the wrong expression for t_2; h/t to James)

- Different observers assign different t_1 and t_2 separately
- But every inertial observer computes the same *product*
- The signs sort the cases: timelike $I < 0$, spacelike $I > 0$, null $I = 0$

Geroch's methodological theme: the structure of relativistic spacetime is represented by the spacetime interval, based on *clocks, light pulses*—and nothing else need be presupposed.

Case 1: Timelike Separation

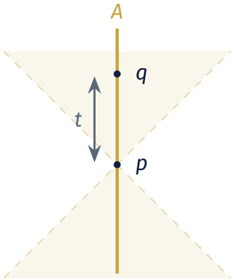
If q lies inside p's future light cone, there is an inertial observer whose worldline passes through p and q. Take A to be that observer.

For this A:

- $\Delta x = 0$ (p and q at the same place)
- $\Delta t = t$ (A's elapsed proper time)
- $t_1 = t, \quad t_2 = -t$

$$I(p,q) = -t^2.$$

A timelike interval encodes **proper time** directly.



Case 2: Spacelike Separation

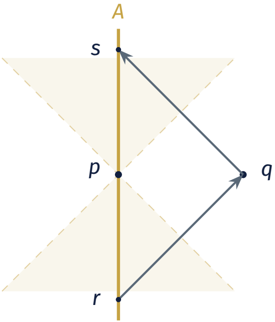
If q lies outside p's light cones, there is no worldline through p and q. *But* there is a unique inertial observer A for whom p and q are simultaneous.

For this A:

- $\Delta t = 0$ (events simultaneous)
- $\Delta x =$ spatial distance in A's frame
- $t_1 = t_2 = \Delta x / c$ (by symmetry)

$$I(p, q) = (\Delta x)^2 / c^2.$$

A spacelike interval encodes **spatial separation** in the simultaneous frame.



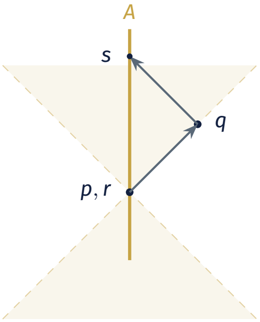
Case 3: Null (Lightlike) Separation

If q lies on p's future light cone, p and q can be joined by a light ray.

Take any inertial observer A through p:

- A emits the signal *at p* itself, so $t_2 = 0$
- The signal reaches q and a return signal arrives at s

$$I(p, q) = t_1 \cdot t_2 = t_1 \cdot 0 = 0.$$



Frame-independence

Every inertial observer computes $I = 0$: the light cone is an absolute feature of spacetime.

The Speed of Light?

The Status of c

Einstein's second postulate took the constancy of light's speed as fundamental. Maudlin (close of Ch. 4) emphasizes that this is misleading.

- Once Newtonian absolute time and space are gone, *nothing* has an objective speed—light included!
- What we do have is the geometric structure: a single null cone at every event, common to all observers

Maudlin's approach

Specify Minkowski geometry directly; connect it to physical behavior via a small set of principles—none of which mentions "the speed of light."

Three Principles in Place of Two Postulates

Replaces Einstein's two postulates with three principles connecting Minkowski geometry to matter:

- **Law of Light**: in vacuum, light propagates along the null worldlines of the metric
- **Relativistic Law of Inertia**: force-free bodies travel along timelike geodesics (straight worldlines of Minkowski spacetime)
- **Clock Hypothesis**: an ideal clock measures the spacetime interval along its worldline

None mentions the speed of anything. The "constancy of c" does not play a role; Maudlin reviews how this can be understood in terms of concrete experimental results.

What Clocks Measure

Two Questions about Clocks

Suppose that we have two clocks A, B constructed in the same way, which pass between two events p and q.

Comparing Clocks

Elapsed times $\Delta t_A = t_A(q) - t_A(p)$ and Δt_B. Compare the readings of clocks in two scenarios – does $\Delta t_A = \Delta t_B$?

- The clocks go along *different* wordlines between p and q.
- The clocks go along wordlines that are as close as possible to each other between p and q.

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- The clocks go along wordlines that are as close as possible to each other between p and q. Yes!

The Clock Hypothesis

This supports the idea that an ideal clock measures an intrinsic geometrical property along its worldline — called **proper time**.

- This is the **clock hypothesis**: clocks are sensitive to the spacetime interval, do not depend on other things (accelerations, past histories, …)
- Empirically, real clocks (atomic, muon decay, etc.) satisfy this to extremely high precision
- The hypothesis is a contingent claim about how physical systems behave

The Twins

The Setup

Alice and Bob are twins. At event E_0, Bob departs Earth in a fast rocket; he travels to a distant star and returns to Alice at event E_1.

- Alice stays on Earth (which we idealize as inertial)
- Bob accelerates outward, coasts, at some point turns around (accelerates), coasts, decelerates
- Both eventually meet again at E_1

Question: when they reunite, who is older?

The Apparent Paradox

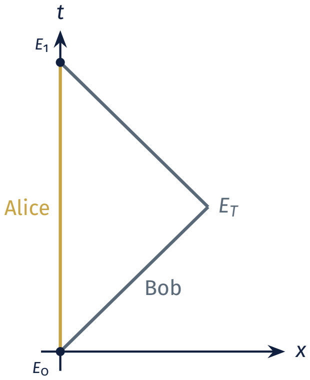
Naively:

- By time dilation, Alice (at rest) sees Bob's clock running slow—so Bob should be *younger* at reunion
- But by the principle of relativity, Bob can equally well regard *himself* as at rest and Alice as moving
- So Bob should see Alice's clock running slow—and *Alice* should be younger
- Both can't be right!

?? *Is special relativity inconsistent? Or is their something faulty in this argument?*

The Minkowski Diagram

- Alice's worldline: a straight vertical line from E_0 to E_1
- Bob's worldline: simplified to a bent path with a turn-around event E_T
- These are two different paths through Minkowski spacetime joining the same pair of events
- Different paths $\rightarrow$ different proper times



Where the Symmetry Breaks

The naive symmetry argument fails because:

- Alice's worldline is **inertial throughout**—a single straight line
- Bob's worldline cannot be a single inertial trajectory: he switches inertial frames at E_T (for this simple picture, but this is also true for more complicated trajectories)
- Relativity of simultaneity: when Bob switches frames at E_T, the events on Earth that count as "now" for him jump forward

Maudlin's framing

The asymmetry between Alice and Bob is *intrinsic* to their worldlines. This is a geometric fact about their worldlines, no appeal to "forces felt during turn-around" is needed.

The Reverse Triangle Inequality

In Euclidean geometry, two sides of a triangle are *longer* than the third.

In Minkowski spacetime

Between two timelike-separated events, the **straight (inertial) worldline** has the *longest* proper time. Any bent path has *less*.

Because Alice's worldline is straight and Bob's is bent, proper time as measured by Alice is *longer* than proper time as measured by Bob. When they reunite, Bob is younger. This is not a paradox; it is geometry.

What the Twins Teach Us

- Time dilation, applied naively to “each twin sees the other’s clock slow,” generates the apparent paradox
- The resolution lies not in any new physics but in the *geometric structure* of Minkowski spacetime
- Proper time depends on path, just as length depends on the path in ordinary geometry
- An inertial worldline is special: it *maximizes* proper time between two events

Summary

1. **Light cones** replace simultaneity slices and define the causal structure of Minkowski spacetime
2. **The interval** Δs^2 is the invariant geometric quantity
3. Proper time along a worldline is what clocks measure
4. **The twins**: not a paradox; the bent worldline has less proper time than the straight one, by the reverse triangle inequality

For Next Time

- **Read:** Maudlin, Ch. 5; Geroch, Chs. 5–6.
- **Think about:**
  - What *is* the proper-time interval, geometrically?
  - In what sense does a clock "measure" it?
  - How does this generalize to non-inertial worldlines?

May 12: **Measurement and the Construction of Minkowski Spacetime**

- Proper time as path length
- Operational vs. geometric readings of SR
- The interval as fundamental

Overflow: The Lorentz Transformations

In coordinates, the boost from S to a frame S' moving with speed v along the x-axis is:

$$t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad x' = \gamma(x - vt), \qquad y' = y, \qquad z' = z.$$

- The factor of vx/c^2 in the time transformation is the formal expression of the relativity of simultaneity
- In the limit $v/c \to 0$: recover the Galilean transformations
- The full symmetry group of Minkowski spacetime is the **Poincaré group** (Lorentz boosts + rotations + translations)

Overflow: Velocity Addition

Galilean velocity addition: $u' = u - v$. Relativistic version:

$$u' = \frac{u - v}{1 - uv/c^2}.$$

- For $u, v \ll c$: recovers the Galilean rule
- For $u = c$: gives $u' = c$ (consistent with the second postulate)
- Two velocities, each less than c, can never combine to give a velocity greater than c

Speed of light is the universal upper bound—a structural feature of Minkowski spacetime, not a contingent property of light itself.

Overflow: Bell's Spaceship Paradox

Two rockets, initially at rest, accelerate identically along the same line. A taut string connects them.

- In the launch frame: the rockets remain the same distance apart (they accelerate identically)
- But the string, in its own (instantaneously co-moving) frame, suffers Lorentz contraction
- The string therefore stretches and eventually snaps
- Each rocket pilot disagrees with the other about "which is moving"—a striking illustration of relativity-of-simultaneity effects

Bell used this case to argue that length contraction is real, not merely apparent.

Overflow: Muon Decay as Empirical Test

Cosmic-ray muons:

- Created in the upper atmosphere ($\sim$ 15 km altitude) when cosmic rays strike air molecules
- Rest-frame lifetime: $\sim 2.2\ \mu$s
- At 0.99c, they would travel only $\sim$ 650 m before decaying—if there were no time dilation
- Observed: large numbers reach the ground
- The dilation factor predicted by SR matches observation precisely

An immediate, everyday confirmation that time dilation is a real geometric feature of the world—not a calculational artifact.

Questions?

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